

# Advanced Topics in Automation and Control Engineering

Aerial Robotics Laboratory

Introduction

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# Laboratory goals

1. Learn theoretical concepts through hands-on approach
  - “DIY” implementations
  - Benchmark against a real implementation
2. Use state-of-the-art software tools and programming languages
  - [Gazebo](#)
  - [telekyb3](#)
  - Python
  - [Docker]
3. Have fun and get passionate about aerial robotics

# Laboratory structure

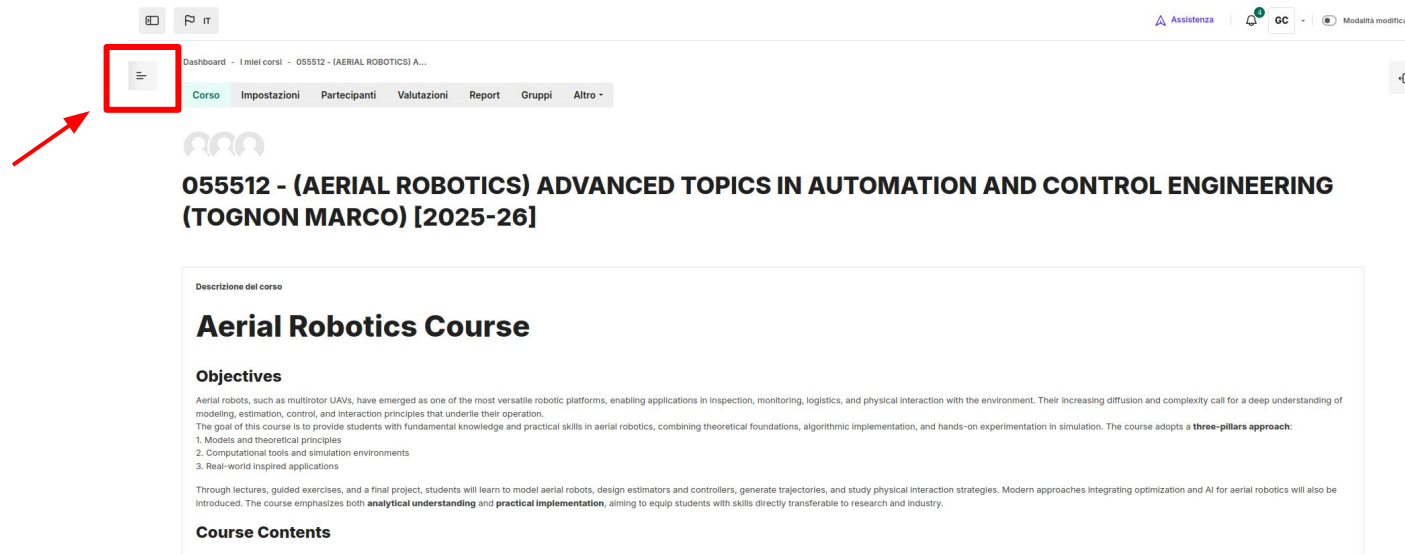
| Week n. | Lectures | Hours | Content                         |
|---------|----------|-------|---------------------------------|
| 5       | 08/04/26 | 3     | Introduction <span>today</span> |
| 5       | 09/04/26 | 2     | Modeling Lecture                |
| 6       | 14/04/26 | 3     | Solutions review + Q&A Modeling |
| 6       | 15/04/26 | 3     | Trajectory generation           |
| 6       | 16/04/26 | 2     | State estimation                |
| 7       | 21/04/26 | 3     | Solutions review + Q&A          |
| 7       | 22/04/26 | 3     | Motion control                  |
| 7       | 23/04/26 | 2     | Physical interaction            |

# Laboratory expected interaction

- In class:
  - a. Presentation topic of the day
  - b. Work in autonomy + Q&A (synchronous interaction)
    - Higher priority to in-person participants
- Outside class / Remote participants:
  - a. Webeep discussions (asynchronous interaction)

# Asynchronous interaction

## Webeep discussions



Dashboard - I miei corsi - 055512 - (AERIAL ROBOTICS) A...

Assistenza GC Modaltà modifica

Corso Impostazioni Partecipanti Valutazioni Report Gruppi Altro

055512 - (AERIAL ROBOTICS) ADVANCED TOPICS IN AUTOMATION AND CONTROL ENGINEERING (TOGNON MARCO) [2025-26]

Descrizione del corso

### Aerial Robotics Course

**Objectives**

Aerial robots, such as multirotor UAVs, have emerged as one of the most versatile robotic platforms, enabling applications in inspection, monitoring, logistics, and physical interaction with the environment. Their increasing diffusion and complexity call for a deep understanding of modeling, estimation, control, and interaction principles that underlie their operation.

The goal of this course is to provide students with fundamental knowledge and practical skills in aerial robotics, combining theoretical foundations, algorithmic implementation, and hands-on experimentation in simulation. The course adopts a **three-pillars approach**:

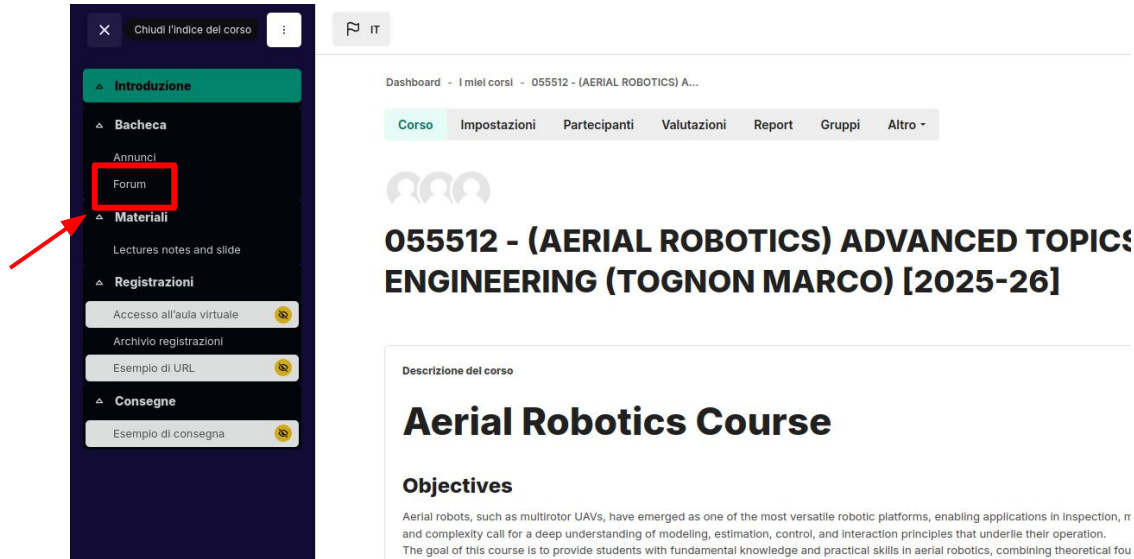
1. Models and theoretical principles
2. Computational tools and simulation environments
3. Real-world inspired applications

Through lectures, guided exercises, and a final project, students will learn to model aerial robots, design estimators and controllers, generate trajectories, and study physical interaction strategies. Modern approaches integrating optimization and AI for aerial robotics will also be introduced. The course emphasizes both **analytical understanding** and **practical implementation**, aiming to equip students with skills directly transferable to research and industry.

**Course Contents**

# Asynchronous interaction

## Webeep discussions



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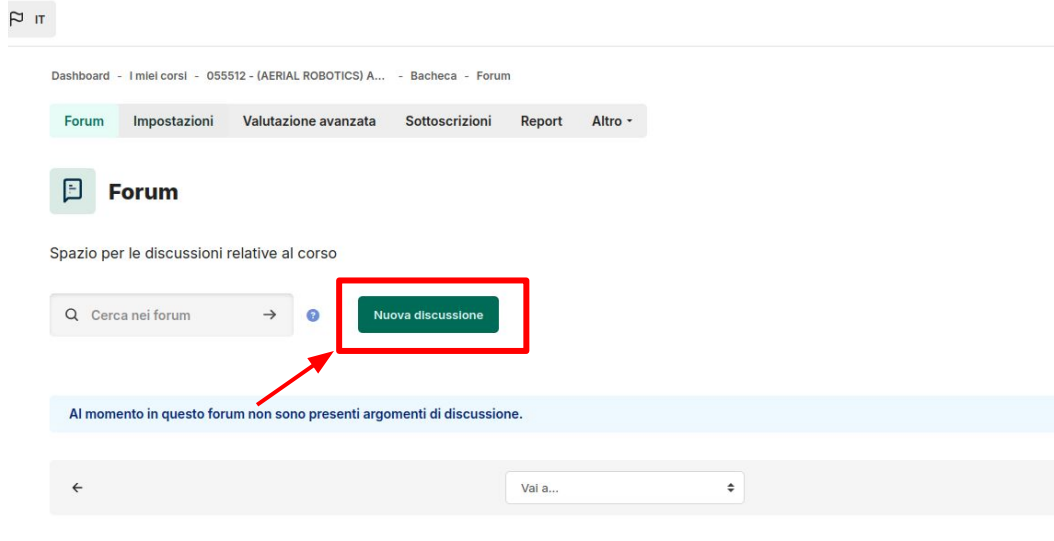
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# Asynchronous interaction

## Webeep discussions



# Asynchronous interaction

## Webeep discussions

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# Asynchronous interaction

Webeep discussions

How to fill it?

I love it!

Discussione



TEMPLATE: [Laboratory]["TOPIC"]["CATEGORY"] "SHORT AND INFORMATIVE TITLE"